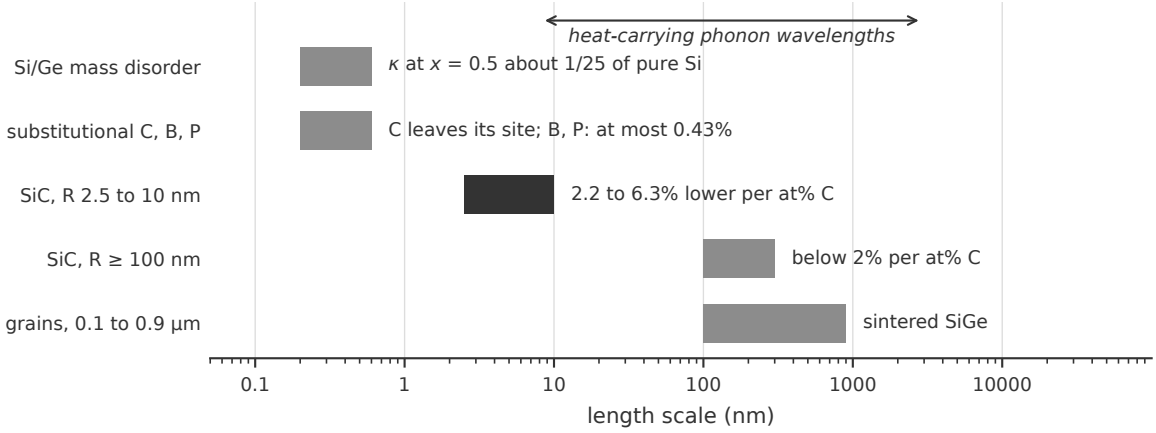
heat-carrying phonon wavelengths
Si/Ge mass disorder
κ at x = 0.5 about 1/25 of pure Si
substitutional C, B, P
C leaves its site; B, P: at most 0.43%
SiC, R 2.5 to 10 nm
2.2 to 6.3% lower per at% C
SiC, R ≥ 100 nm
below 2% per at% C
grains, 0.1 to 0.9 µm
sintered SiGe
0.1
1
10
100
1000
10000
length scale (nm)